

Notes: Straub and Ulbricht (RES, 2024)

Endogenous Uncertainty and Credit Crunches

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1 Big picture

- **Question.** Can a temporary financial shock create a *persistent* recession because it endogenously raises uncertainty, and if so, what is the mechanism?
- **Core idea.** Firms need outside funding to operate a risky high-productivity technology. Whether they get funding depends on banks' beliefs about their potential. If a firm is denied funding, it operates a safe low-productivity technology and generates less information about its true potential. Less information raises uncertainty, and higher uncertainty further tightens credit.
- **Main mechanism.** There is a two-way feedback loop between credit and learning: tighter credit reduces information production, and worse information raises credit spreads and denial rates. A short-lived shock can therefore become internally persistent.
- **Main theoretical result.** Endogenous uncertainty creates amplification and persistence that are absent in a comparable model where uncertainty is fixed exogenously.
- **Main quantitative result.** A financial shock with a half-life of four quarters generates an output response with a half-life of about sixteen quarters once endogenous uncertainty is active; without it, the output half-life is again four quarters.
- **Empirical relevance.** The model lines up with familiar crisis facts: persistent drops in output and employment, higher credit spreads, more defaults, more dispersion in firm sales, and more disagreement among forecasters.
- **Why this paper matters.** It gives a micro-founded propagation mechanism for credit crises that works through *firm-level information loss*. It also connects naturally to macro wedges, so the same force can be embedded in RBC and New Keynesian environments.

2 Data and measurement (key objects)

2.1 Model economy and unit of analysis

- **Unit of analysis.** A continuum of heterogeneous firms indexed by $i \in [0, 1]$.
- **Firm-level state.** Each firm has risky productivity $A_{i,t}$, beliefs about that productivity $(\mu_{i,t}, \Sigma_{i,t})$, and a bankruptcy flag $d_{i,t}$.
- **Aggregate objects.** Output Y_t , employment L_t , wage w_t , lending threshold ν_t , and in the capital extension the aggregate capital stock K_t .
- **Interpretation.** The model is theory-first, but it is disciplined by empirical targets and by moments from U.S. business-cycle data.

2.2 Unobservables and timing

- **Risky productivity.** The firm's potential technology is summarized by $A_{i,t}$, which is not observed at the time the technology and financing decisions are made.
- **Beliefs.** Agents know only the posterior mean $\mu_{i,t}$ and posterior variance $\Sigma_{i,t}$ of $\log A_{i,t}$.

- **Timing within a period.**

- Morning: bankruptcy flags may be removed; firms choose baseline or risky technology; banks decide whether to fund the required operating cost.
- Afternoon: some productivity realizations are revealed; firms hire labor, produce, repay loans if possible, and may default.

- **Key information friction.** Firms that operate the risky technology reveal their productivity. Firms that do not operate it usually do *not* reveal it, except through an exogenous information signal that arrives with probability θ .

2.3 Calibration targets and quantitative inputs

- **Revenue productivity process.** Persistence ρ and innovation volatility σ_ε are chosen to match the revenue-productivity process estimated by Foster, Haltiwanger, and Syverson.
- **Effect of losing credit.** The low-productivity baseline technology $\tilde{A}$ is chosen so that losing risky credit implies about a 30% marginal employment drop, guided by Chodorow-Reich’s evidence.
- **Steady-state denial rate.** The operating cost ϕ is chosen so that about 25% of firms lack risky funding in steady state, implying a sales-weighted non-dividend share of about 8%.
- **Bankruptcy and information duration.** The bankruptcy-removal rate η targets an average bankruptcy duration of 2.86 quarters, and the exogenous information-revelation rate θ targets an average risky-funding denial duration of about 2 years.
- **Macro estimation in Section 6.** The quantitative exploration estimates shock processes using U.S. data on real GDP, investment, consumption, non-farm payrolls, and non-financial business lending transactions.

2.4 Empirical applications in the paper

- Partial-equilibrium dynamics of a single firm after a credit disruption.
- General-equilibrium impulse responses to an aggregate financial shock.
- RBC and New Keynesian embeddings of the same micro mechanism.
- Bayesian estimation of a richer model to U.S. business-cycle data and an interpretation of the Great Recession.
- A supplementary comparison between model-implied forecaster disagreement and survey data on professional forecasters.

3 Models and empirical specifications (all equations + interpretation)

3.1 Final-good aggregation and firm technology

A competitive final-good producer aggregates differentiated intermediate goods:

$$Y_t = \left(\int_0^1 Y_{i,t}^{\frac{\xi-1}{\xi}} di \right)^{\frac{\xi}{\xi-1}},$$

where $\xi > 1$ is the elasticity of substitution across varieties. Final-good demand for firm i is

$$Y_{i,t} = Y_t p_{i,t}^{-\xi}. \quad (1)$$

Each intermediate producer has linear technology

$$Y_{i,t} = A_{i,t}^{\frac{1}{\xi-1}} L_{i,t}. \quad (2)$$

The risky productivity evolves as

$$\log A_{i,t} = \rho \log A_{i,t-1} + (1 - \rho) \log \bar{A} + \varepsilon_{i,t}, \quad (3)$$

with innovation variance σ_ε^2 . There is also a baseline technology with constant productivity $\bar{A}$.

- **Interpretation.** The risky technology is a firm's *potential*. It is better on average, but the firm must commit before knowing the current realization of $A_{i,t}$.
- **Why the exponent matters.** The exponent $1/(\xi - 1)$ is chosen so that $A_{i,t}$ maps directly into revenue productivity, which simplifies both the firm problem and the aggregate wedges later on.

3.2 Financial friction and costly external funding

Firm i chooses between the baseline and risky technologies in the morning. The baseline technology requires a fixed operating cost $\tilde{\phi}$, while the risky technology requires a larger fixed operating cost $\phi > \tilde{\phi} > 0$. Firms cannot self-finance these costs and must borrow from banks.

Operating profits are

$$\Pi_{i,t} \equiv p_{i,t} Y_{i,t} - w_t L_{i,t}. \quad (4)$$

Default occurs if profits fall short of required repayment:

$$\Pi_{i,t} < (1 + r_{i,t}) \phi_{i,t}. \quad (5)$$

Banks lend competitively and choose an interest rate satisfying a zero-profit condition:

$$(1 + r_{i,t}) \left(1 - \Pr_t [\Pi_{i,t} < (1 + r_{i,t}) \phi_{i,t}] \right) = 1 + \lambda_t. \quad (6)$$

- **Interpretation.** λ_t is a financial shock to banks' cost of funds. A higher λ_t tightens lending conditions economy-wide.
- **Default consequences.** A defaulting firm receives a bankruptcy flag and loses access to risky funding until the flag is removed, which happens each period with probability η .
- **Key force.** Banks care about both pessimism and uncertainty because both raise default risk and therefore increase loan spreads.

3.3 Households and aggregate demand in the baseline

The representative household maximizes

$$\mathbb{E}_t \sum_{\tau=t}^{\infty} \beta^{\tau-t} \left[\log C_t - v \frac{1}{1+\zeta^{-1}} L_t^{1+\zeta^{-1}} \right], \quad (7)$$

subject to the flow budget constraint

$$C_t + B_{t+1} = w_t L_t + (1 + r_{t-1}) B_t + T_t. \quad (8)$$

- **Interpretation.** The household side is standard. In the baseline model, persistence does not come from household habits or sticky prices but from the information-credit feedback at the firm level.

3.4 Information friction and belief dynamics

The crucial information friction is that current risky productivity is not always observed. If a firm operates the risky technology, or if an exogenous information signal arrives, then the current realization becomes known. Otherwise, agents do not learn the period- t risky productivity.

Beliefs are summarized by posterior mean and variance:

$$\mu_{i,t} \equiv \mathbb{E}_t[\log A_{i,t} \mid I_t], \quad \Sigma_{i,t} \equiv \text{Var}_t[\log A_{i,t} \mid I_t].$$

They evolve according to

$$\mu_{i,t+1} = \begin{cases} \rho \log A_{i,t} + (1 - \rho) \log \bar{A}, & i \in B_t, \\ \rho \mu_{i,t} + (1 - \rho) \log \bar{A}, & i \notin B_t, \end{cases} \quad (14)$$

$$\Sigma_{i,t+1} = \begin{cases} \sigma_\varepsilon^2, & i \in B_t, \\ \rho^2 \Sigma_{i,t} + \sigma_\varepsilon^2, & i \notin B_t, \end{cases} \quad (15)$$

where B_t is the set of firms whose risky productivity is observed at date t .

- **Interpretation.** A funded firm keeps learning and uncertainty resets to the innovation variance. An unfunded firm stops generating information, so uncertainty accumulates over time.
- **This is the model's central dynamic channel.**

3.5 Equilibrium characterization and the lending threshold

Profit maximization under monopolistic competition implies a constant-markup pricing rule and profits proportional to productivity. Combining this with banks' zero-profit condition yields a cutoff rule for risky lending.

Proposition 1. Define the lending threshold

$$\nu_t \equiv \log((1 + \lambda_t)\phi) - \log\left(\frac{Y_t}{w_t^{\xi-1}}\right) + \log(\xi^\xi (\xi - 1)^{1-\xi}).$$

Then a firm obtains risky funding if and only if it has no bankruptcy flag and

$$\mu_{i,t} - V(\Sigma_{i,t}) \geq \nu_t, \quad (13)$$

where

$$V(\Sigma) \equiv \min_{x \in (0,1]} \left\{ \Phi^{-1}(x) \sqrt{\Sigma} - \log x \right\}.$$

- **Interpretation.** $\mu_{i,t}$ raises credit access, while $V(\Sigma_{i,t})$ is an uncertainty penalty. A more uncertain firm needs a higher expected productivity to get funded.
- **Role of aggregates.** The threshold ν_t depends on the financial shock λ_t , aggregate output Y_t , and the wage w_t . So firm-level beliefs and aggregate conditions interact in equilibrium.

Aggregating across firms delivers an efficiency wedge:

$$A_t \equiv \left(\int_0^1 A_{i,t} di \right)^{\frac{1}{\xi-1}}, \quad (17)$$

and equilibrium aggregate relationships

$$w_t = (1 - \xi^{-1})A_t, \quad Y_t = A_t L_t. \quad (16)$$

Combining these with labor supply gives

$$(1 - \xi^{-1})A_t = v L_t^{1/\zeta} \left(A_t L_t - \int_0^1 \phi_{i,t} di \right). \quad (18)$$

- **Interpretation.** Financial disruptions worsen aggregate efficiency because more firms are pushed onto the low-productivity baseline technology.
- **Uniqueness.** The paper shows that, for $\xi > 2$, the general equilibrium fixed point is unique.

3.6 Interaction between credit and learning

The core firm-level constraint can be written as

$$\mu_{i,t} - V(\Sigma_{i,t}) < \nu_t. \quad (19)$$

With no shocks and no exogenous revelation, the joint dynamics of $(\mu_{i,t}, \Sigma_{i,t})$ can be visualized in a phase diagram.

Proposition 2. There exist two thresholds $\underline{\nu}$ and $\bar{\nu}$ such that if

$$\underline{\nu} \leq \nu \leq \bar{\nu},$$

there are two non-stochastic firm-level steady states; otherwise the steady state is unique.

- **Interpretation.** The model can generate local hysteresis. For intermediate lending thresholds, a firm can get stuck in a low-information, no-funding state even though fundamentals themselves are unchanged.
- **Economic intuition.** If a firm loses funding, uncertainty rises. If uncertainty rises enough before mean beliefs recover, the firm stays constrained even after the original shock disappears.

A temporary disruption in credit can therefore create *internal persistence*: output falls immediately because the firm switches to the baseline technology, but the main difference from the exogenous-uncertainty benchmark shows up after the financial shock itself is gone.

3.7 Informational externalities and social gains from lending

The paper shows that lending is inefficient for two reasons.

- **Static inefficiency.** Because default is costly, private lending is below the static social optimum.
- **Dynamic inefficiency.** Banks do not internalize the option value of learning. Funding a firm today produces information about its risky technology that improves future lending decisions.

The private willingness to lend is

$$\phi_{i,t}^{\text{private}} = \phi \exp\{\mu_{i,t} - V(\sigma_\varepsilon^2) - \nu\},$$

while the dynamic social benefit of risky lending exceeds the static benefit precisely around the lending threshold.

- **Interpretation.** Lending to a marginal firm is socially valuable because it reveals whether that firm should be funded in the future. This is why public support can crowd *in* future private lending rather than just replacing it contemporaneously.

3.8 Aggregate credit crunches and baseline parameterization

The economy is initialized at its stochastic steady state and is then hit by an aggregate financial shock λ_t that follows an AR(1) process with half-life four quarters. The shock magnitude is normalized to create a 1% drop in output on impact.

The baseline calibration in Table 1 is:

$$\begin{aligned} \beta &= 0.99, & \zeta &= 2, & \xi &= 5, \\ \rho &= 0.944, & \sigma_\varepsilon &= 0.073, & \eta &= 0.350, & \theta &= 0.117, \end{aligned}$$

with

$$\tilde{A}/\bar{A} = 0.563, \quad \phi = 0.139$$

in the endogenous-uncertainty economy.

- **Main quantitative result.** With endogenous uncertainty, output recovery is much slower than in the exogenous-uncertainty counterfactual: the output half-life is sixteen quarters rather than four.
- **Mechanism in the impulse responses.** At date 0 the shock works through tighter credit conditions alone because uncertainty is predetermined. Starting at date 1, the rise in uncertainty amplifies and prolongs the recession.

The model also predicts a rise in credit spreads, default rates, cross-sectional sales dispersion, and disagreement among outside forecasters. If forecaster j observes a private signal

$$\omega_{ij,t} = \log A_{i,t} + \psi_{ij,t},$$

then disagreement is

$$\text{sd}_j[\tilde{\mu}_{ij,t}] = \frac{\sigma_\psi^{-1}}{\Sigma_{i,t}^{-1} + \sigma_\psi^{-2}}. \quad (21)$$

Hence disagreement is increasing in the firm's uncertainty and therefore in the degree to which it is financially constrained.

3.9 Extension with capital: RBC equivalence

When capital is added, firm production becomes

$$Y_{i,t} = A_{i,t}^{\frac{1}{\xi-1}} K_{i,t}^\alpha L_{i,t}^{1-\alpha}.$$

The lending threshold becomes

$$\nu_t = \log((1 + \lambda_t)\phi) - \log\left(\frac{Y_t}{(1 + r_t^K)^{\alpha(\xi-1)} w_t^{(1-\alpha)(\xi-1)}}\right) + \log(\xi^\xi (\xi - 1)^{1-\xi}).$$

Proposition 3. Conditional on the induced efficiency wedge A_t and resource wedge

$$G_t \equiv \int_0^1 \phi_{i,t} di, \quad (22)$$

aggregate quantities satisfy a standard RBC system:

$$C_t^{-1} = \mathbb{E}_t[\beta((1 - \xi^{-1})\alpha A_{t+1} K_{t+1}^{\alpha-1} L_{t+1}^{1-\alpha} + 1 - \delta) C_{t+1}^{-1}], \quad (23)$$

$$v L_t^{1/\zeta} = (1 - \xi^{-1})(1 - \alpha) C_t^{-1} A_t K_t^\alpha L_t^{-\alpha}, \quad (24)$$

$$C_t + G_t + K_{t+1} = A_t K_t^\alpha L_t^{1-\alpha} + (1 - \delta) K_t. \quad (25)$$

- **Interpretation.** The micro credit-learning friction maps into endogenous RBC wedges.
- **Why persistence increases.** A tighter lending threshold lowers the efficiency wedge, which depresses investment. Lower future capital then further tightens credit conditions, adding another feedback loop.

3.10 New Keynesian extension and equity-financed firms

In the nominal-rigidity extension, half of households are hand-to-mouth and nominal wages are rigid. Aggregate consumption is

$$C_t = C_t^{\text{base}} + \frac{w_t L_t}{2},$$

with

$$(C_t^{\text{base}})^{-1} = \mathbb{E}_t[\beta(1 + r_{ss})(C_{t+1}^{\text{base}})^{-1}].$$

- **Interpretation.** Financial shocks now propagate through aggregate demand as well as supply. Lower lending reduces firms' spending and income, hand-to-mouth households cut consumption, and the economy exhibits a persistent positive labour wedge.

The paper also studies a variant in which half of firms are equity-financed and therefore do not need bank credit to pay the fixed operating cost. Those firms always operate their preferred technology and do not suffer from the information friction.

- **Interpretation.** This extension scales down the impact effect but leaves persistence largely intact. The mechanism does not rely on every firm being bank-dependent.

3.11 Quantitative exploration and aggregate-versus-firm uncertainty

The richer model in Section 6 adds four standard aggregate shocks in addition to the financial shock: a TFP shock Z_t , a labour wedge shock τ_t^L , an investment wedge shock τ_t^I , and a government spending shock τ_t^G . The five shock processes are estimated by Bayesian methods.

A key result is that endogenous uncertainty matters mainly for *financial* shocks. It changes little about the propagation of non-financial shocks.

To study whether uncertainty must be firm-level, the paper considers an alternative in which all uncertainty is about aggregate productivity Z_t and agents infer it from a noisy signal of past aggregate output:

$$s_t = Y_{t-1} + \omega_t = Z_{t-1}X_{t-1} + \omega_t. \quad (27)$$

Aggregate uncertainty then follows

$$\Sigma_t^Z = \frac{\rho_Z^2 \Sigma_{t-1}^Z}{1 + (\sigma_\omega / X_{t-1})^{-2} \Sigma_{t-1}^Z} + \sigma_Z^2. \quad (28)$$

- **Interpretation.** This version produces almost no amplification. For a 1% impact output loss, peak aggregate uncertainty rises by less than 0.6%, far below the roughly 10% rise in average firm-level uncertainty in the baseline model.
- **Main message.** The paper’s mechanism works because learning breaks down at the *firm level*. Aggregate uncertainty does not move enough because aggregate signals remain too precise, even in a crisis.

4 Identification and instruments (what makes the estimates credible)

4.1 No external instrument here: credibility comes from a sharply disciplined mechanism

This is not an instrumental-variables paper. The credibility of the paper comes instead from a tight mapping between a transparent mechanism and a broad set of qualitative and quantitative facts.

- The mechanism requires firms to depend on outside funding for risky activity.
- It requires learning to be state contingent: no risky activity means little or no new information.
- It then delivers a very specific propagation pattern: financial shocks become more persistent than non-financial shocks.

4.2 What disciplines the model quantitatively

- **Productivity dynamics.** The persistence and volatility of firm-level productivity are imported from external evidence.
- **Employment response to losing credit.** The productivity gap between risky and baseline technologies is pinned down to match evidence on the employment effect of losing credit access.

- **Steady-state funding denial.** The fixed operating cost is calibrated to match plausible denial rates from firm-level financing evidence.
- **Persistence of denial and bankruptcy.** The parameters θ and η are tied to observed denial duration and bankruptcy duration.
- **Business-cycle estimation.** In the richer model, shock processes are estimated against macro aggregates and lending data rather than chosen ad hoc.

4.3 Why the exogenous-uncertainty counterfactual is so useful

A major strength of the paper is that it repeatedly compares the full model to a counterfactual in which uncertainty is held fixed at its lower bound.

- This isolates what is coming specifically from the endogenous uncertainty channel.
- On impact, both models look similar, because uncertainty is predetermined.
- The gap opens only after the shock, exactly where the theory says the feedback loop should matter.

4.4 Why the firm-level uncertainty assumption is essential

The aggregate-uncertainty alternative is an important discipline device.

- If the relevant uncertainty were only aggregate, the model would generate almost no amplification.
- The paper shows this directly: aggregate uncertainty barely moves even under the most favourable parameterization.
- So the persistence result is not just “uncertainty matters” in the abstract. It is specifically the interaction between credit and *firm-level* information loss that matters.

4.5 Relation to the literature

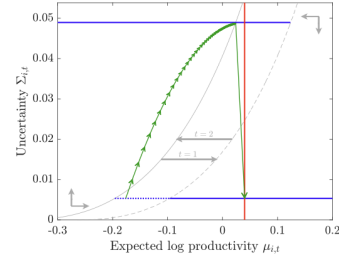
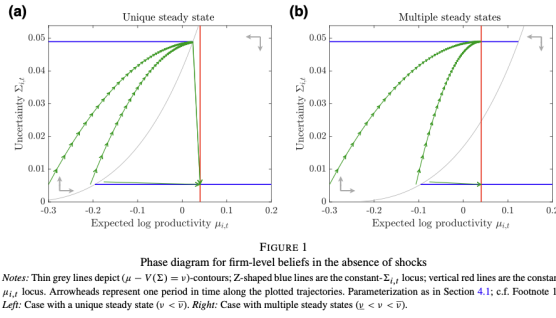
- Relative to exogenous uncertainty papers, this paper endogenizes uncertainty through the production of information.
- Relative to the dispersed-information macro literature, it ties information frictions directly to credit-market access.
- Relative to financial-frictions models with wedges, it provides a micro foundation for persistent efficiency and labour wedges after financial shocks.

5 Tables and figures

5.1 Figure 1: Phase diagram for firm-level beliefs

Read:

- The figure is the cleanest picture of the mechanism. The grey contour is the risky-funding cutoff, the red vertical line is the constant-expectations locus, and the blue Z-shaped object is the constant-uncertainty locus.
- In the left panel there is a unique steady state, but credit disruptions can still be persistent because firms move slowly through the phase diagram.
- In the right panel, for intermediate lending thresholds, there are multiple steady states: the same firm can be trapped in a no-funding, high-uncertainty state.
- This figure shows that the model's persistence is internal. It does not require a permanently bad exogenous shock.



5.2 Figures 2 and 3: Temporary credit disruption and firm-level dynamics

Read:

- Figure 2 shows how a one-period worsening in credit conditions can push a marginal firm across the original funding contour so that the recovery of the financial shock does not restore risky funding.
- Figure 3 then simulates that logic in the full stochastic model: after the one-period shock, the firm in the endogenous-uncertainty model remains constrained, while the exogenous-uncertainty counterfactual recovers almost immediately.
- The main firm-level symptoms are persistent low employment, persistently depressed expected log productivity, and a hump-shaped rise in uncertainty that peaks after the initial shock.
- The figure connects the phase-diagram logic directly to the empirical evidence that temporary credit disruptions can have multi-year effects.

5.3 Figure 4: Private and social benefits from lending

Read:

- The private willingness to fund is below the static social benefit for all firms around the threshold because default risk creates a standard lending wedge.
- The dynamic social benefit spikes sharply near the lending threshold, where funding today is especially informative about future creditworthiness.

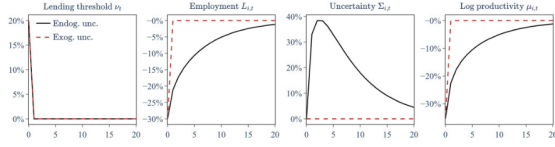


FIGURE 3
Impact of temporary financial shock on firm dynamics

Notes: Solid line: Effect of one time disruption in credit, $v_0 > v$, in period $t = 0$ on the average evolution of a firm close to the funding threshold, $\rho \log A = v + V(\sigma_\varepsilon^2)$. Dashed line: Same evolution, but fixing uncertainty exogenously at $\Sigma_{i,t} = \sigma_\varepsilon^2$. Parameterization as in Section 4.1.

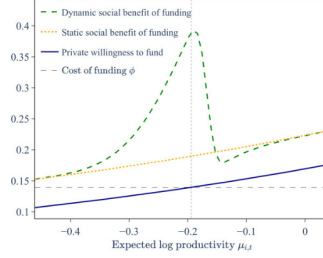


FIGURE 4
Private and social benefits from lending

Notes: The vertical dotted line indicates the equilibrium threshold $v + V(\sigma_\varepsilon^2)$ for lending. Parameterization as in Section 4.1.

- This is the paper's core welfare result: public lending has value not only because it keeps firms operating today, but because it creates information that improves future private lending.

5.4 Table 1: Calibrated parameters

Read:

- The key baseline parameters are $\rho = 0.944$, $\sigma_\varepsilon = 0.073$, $\phi = 0.139$, $\eta = 0.350$, and $\theta = 0.117$ in the endogenous-uncertainty model.
- The exogenous-uncertainty counterfactual differs mainly in setting $\theta = 1$ and recalibrating ϕ so that the impact response of output to a financial shock is the same.
- This makes the comparison conservative: persistence differences are not mechanical consequences of different impact sizes.

Parameter	β	ζ	ζ'	v	$\bar{A}$	$\bar{A}/\bar{A}$	ρ	σ_ε	ϕ	η	θ
Endog. uncertainty	0.99	2.000	5.000	0.913	1.041	0.563	0.944	0.073	0.139	0.350	0.117
Exog. uncertainty	0.99	2.000	5.000	0.907	0.983	0.563	0.944	0.073	0.121	0.350	1.000

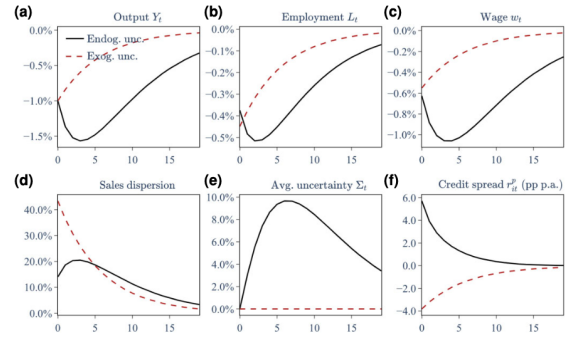


FIGURE 5
General equilibrium response to an AR(1) financial shock

Notes: (a)–(e) are denominated in percentage deviations from the steady state; (f) shows the increase in credit spreads of the 10% least productive firms that still received funding relative to the steady state denominated in percentage points per annum. Solid black lines are model responses; dashed red lines are exogenous uncertainty counterfactuals where uncertainty is fixed at $\Sigma_{i,t} = \sigma_\varepsilon^2$ for all i and t .

5.5 Figures 5 and 6: Aggregate credit crunch and the distribution of uncertainty

Read:

- Figure 5 shows the baseline aggregate responses. Output, employment, and wages fall more and recover much more slowly under endogenous uncertainty than under exogenous uncertainty.
- At quarter 4, output in the endogenous-uncertainty economy is still more than 1.5% below steady state, while the exogenous-uncertainty economy has already recovered half of its impact loss.

- Average uncertainty rises to roughly 10%, sales dispersion increases, and the credit spread of the 10% weakest funded firms rises by more than 4 percentage points.
- Figure 6 explains where the uncertainty comes from: by five periods into the shock, many more firms are stuck in states with several consecutive periods without new information.

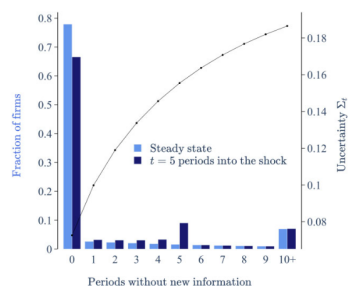


FIGURE 6
The distribution of uncertainty five periods into the impulse response and at the steady state

Notes: Solid black line: level of uncertainty corresponding to s periods without new information (right axis).

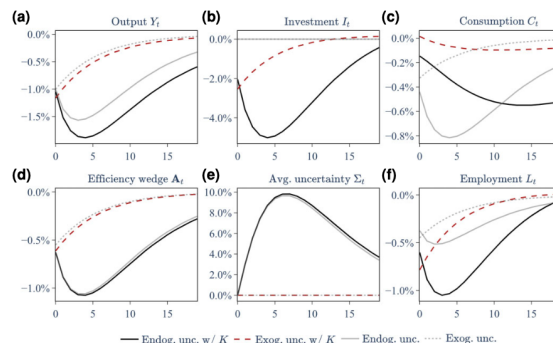


FIGURE 7
General equilibrium response in the model with capital

5.6 Figure 7: Model with capital

Read:

- Adding capital makes the responses more persistent, not less.
- The crucial reason is investment: the financial shock lowers the efficiency wedge, which sharply depresses investment, and lower investment tightens future credit conditions further.
- This is the paper's bridge from a firm-level information story to persistent macro dynamics in an RBC environment.

5.7 Figures 8 and 9: Nominal rigidities and equity-financed firms

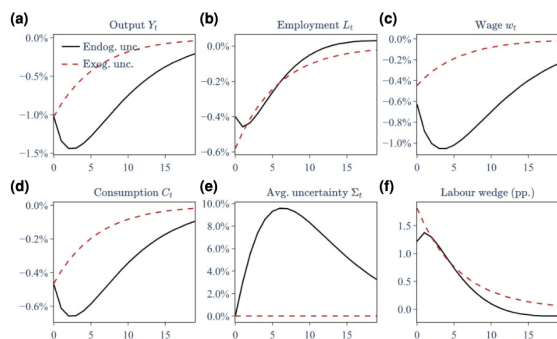


FIGURE 8
Response to financial shock with nominal rigidities and hand-to-mouth agents

Notes: All parameters as in Section 4.1. Share of hand-to-mouth agents of 50%.

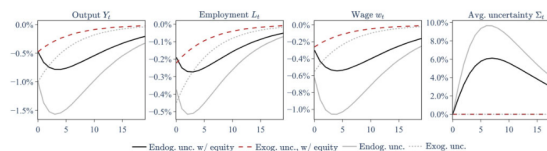


FIGURE 9
General equilibrium response in the model with both bank-financed and equity-financed firms

Read:

- Figure 8 shows that the mechanism survives in a New Keynesian environment. The recession becomes demand driven, and a positive labour wedge emerges, but the persistence logic is still endogenous uncertainty.
- Figure 9 shows that the mechanism also survives when half the firms are equity financed. The impact effect is smaller, but persistence remains very similar.
- Together, these figures make the paper’s point that the credit-learning channel is robust to richer macro environments and to heterogeneity in financing structure.

5.8 Tables 2 and 3: Bayesian estimation and business-cycle moments

TABLE 2
Priors and posteriors

Shock		Prior distribution	Posterior	
			Mode	std. dev.
TFP shock $\log Z_t$	s.d.	Invgamma(0.1, 2)	0.367	(0.018)
	AR	Beta(0.5, 0.1)	0.246	(0.001)
Labour wedge shock τ_t^L	s.d.	Invgamma(0.1, 2)	0.474	(0.023)
	AR	Beta(0.5, 0.1)	0.974	(0.006)
Investment shock τ_t^I	s.d.	Invgamma(0.1, 2)	0.159	(0.009)
	AR	Beta(0.5, 0.1)	0.836	(0.027)
Gov. spending shock τ_t^G	s.d.	Invgamma(0.1, 2)	0.306	(0.015)
	AR	Beta(0.5, 0.1)	0.935	(0.009)
Financial shock λ_t	s.d.	Invgamma(0.1, 2)	4.472	(0.218)
	AR	Beta(0.5, 0.1)	0.821	(0.028)

Notes: Magnitudes are in percentage points. The magnitude of τ_t^G is in percentage points of steady state output.

TABLE 3
Business cycle moments with and without endogenous uncertainty

	Financial shocks		Non-financial shocks	
	Endog. unc.	Exog. unc.	Endog. unc.	Exog. unc.
Output				
std. dev.	0.83	0.45	1.64	1.40
autocorr, lag 1	0.93	0.88	0.93	0.92
autocorr, lag 2	0.74	0.56	0.75	0.70
autocorr, lag 3	0.49	0.20	0.50	0.42
Hours				
std. dev.	0.50	0.30	1.52	1.24
autocorr, lag 1	0.93	0.87	0.94	0.94
autocorr, lag 2	0.74	0.56	0.79	0.78
autocorr, lag 3	0.48	0.19	0.57	0.57
Investment				
std. dev.	2.33	0.97	5.14	4.39
autocorr, lag 1	0.94	0.87	0.92	0.90
autocorr, lag 2	0.77	0.56	0.70	0.64
autocorr, lag 3	0.53	0.19	0.42	0.32
Consumption				
std. dev.	0.12	0.03	0.86	0.84
autocorr, lag 1	0.96	0.95	0.94	0.93
autocorr, lag 2	0.86	0.80	0.77	0.76
autocorr, lag 3	0.72	0.59	0.54	0.52

Notes: Columns 2 and 3 show the volatility and persistence of output and employment associated with financial shocks. Columns 4 and 5 show the volatility and persistence of output and employment associated with the four non-financial shocks (TFP, labour wedge, investment wedge, and government shock). Units of the standard deviations are percentage points relative to the long-run trend.

Read:

- Table 2 reports the estimated shock processes. The financial shock is large and persistent, with posterior standard deviation 4.472 percentage points and AR coefficient 0.821.
- Table 3 is the key business-cycle comparison. Financial shocks generate much more volatility and persistence with endogenous uncertainty than without it: output volatility is 0.83 versus 0.45, and the lag-3 autocorrelation of output is 0.49 versus 0.20.
- By contrast, non-financial shocks look quite similar across the two models. This is critical evidence that the mechanism is about *financial* shocks specifically.

5.9 Figure 10: Endogenous uncertainty and the Great Recession

Read:

- The black line is the data, matched exactly by the full endogenous-uncertainty model. The dashed line is the exogenous-uncertainty counterfactual fed with the same shocks.
- Without endogenous uncertainty, the peak-to-trough drop in output during the Great Recession would have been about half as large, and the economy would have recovered much earlier.
- This is the most policy-relevant figure in the paper: the mechanism is not just visible in stylized impulse responses, but materially changes the interpretation of a major historical event.

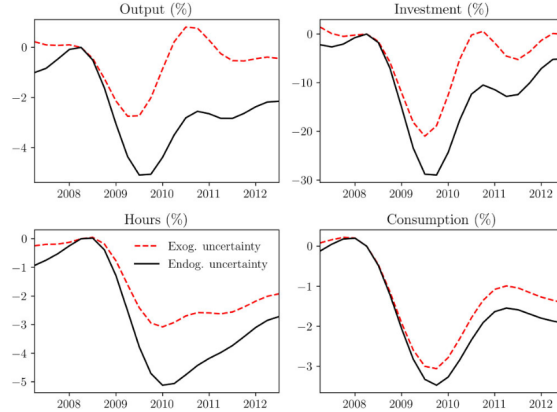


FIGURE 10
Contribution of endogenous uncertainty to the Great Financial Crisis

Notes: This plot illustrates the role of endogenous uncertainty for the behaviour of output, hours, investment, and consumption around the Great Financial Crisis. The solid line is the data, which the endogenous uncertainty model matches exactly. The dashed line simulates the exogenous uncertainty model, subject to the same shock realizations as the endogenous uncertainty model. Both plots are normalized to 0 in 2008Q1. Units are percentage points relative to the long-run trend.

5.10 Figures 11 and 12: Firm-level versus aggregate uncertainty

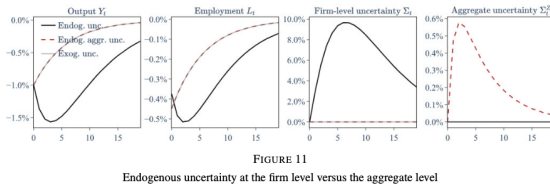


FIGURE 11
Endogenous uncertainty at the firm level versus the aggregate level

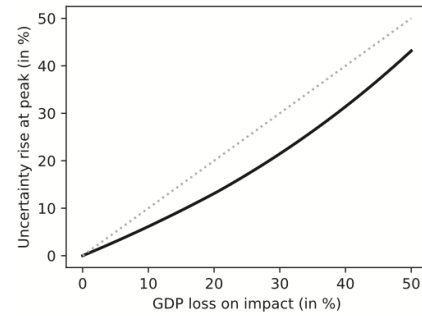


FIGURE 12
Peak increase in uncertainty by size of the financial shock in the aggregate uncertainty model

Read:

- Figure 11 shows that the aggregate-uncertainty model is almost indistinguishable from the exogenous-uncertainty benchmark. For a 1% impact output loss, peak aggregate uncertainty rises by less than 0.6%.
- Figure 12 scales up the shock and shows that even for very large crises, the increase in aggregate uncertainty stays proportionally smaller than the output loss.
- These figures are crucial for interpretation: the paper's mechanism is fundamentally about *firm-level* information loss, not about a generic aggregate uncertainty channel.

6 Short summary

- **Conclusion.** Temporary financial shocks can create persistent recessions because they reduce the information generated about firms' fundamentals, and that loss of information tightens future credit conditions.

- **Main theoretical insight.** Credit and learning interact in a self-reinforcing way. A firm denied risky funding becomes more uncertain, and a more uncertain firm is harder to fund.
- **Main quantitative lesson.** Endogenous uncertainty matters primarily for financial shocks. It greatly amplifies and prolongs them, but has little effect on the propagation of standard non-financial shocks.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes a mechanism through which a short-lived financial disturbance can generate a long-lived recession.
- The persistence does not come from exogenously persistent uncertainty, habits, or nominal rigidities alone; it comes from endogenously impaired learning at the firm level.
- The same mechanism can be embedded in RBC and New Keynesian models and appears in the data chiefly through the propagation of financial shocks.

7.2 Economic intuition

A financially constrained firm does not just produce less today. It also reveals less about what it is capable of doing. That informational loss makes lenders more cautious tomorrow, raising spreads and lowering the chance that the firm regains access to risky funding. Because the firm-level learning problem is repeated across many firms at once in a crisis, a temporary aggregate financial shock can create a persistent aggregate downturn.

The model therefore turns uncertainty from a passive by-product of crises into an active state variable produced by the crisis itself. That is the paper's main conceptual contribution.

7.3 Takeaway

This paper is best understood as a micro-founded theory of why credit crunches can outlast the shocks that trigger them. Its main message is that information destruction is a quantitatively important part of financial crises. If firms cannot operate at potential, the economy stops learning which firms are good risks. That informational scarring slows recovery, makes recessions deeper, and opens up a concrete policy possibility: direct support to firms can be valuable not only because it sustains activity today, but because it preserves the information needed for markets to lend efficiently tomorrow.